

AI Performance Report

Test: demo-checkout-load-2.0.3 Environment: staging Application: smart-shop Started: 2026-07-26 06:00:18 UTC
Stopped: 2026-07-26 06:30:18 UTC Generated: 2026-07-26 06:30:18 UTC

24871

Samples

148

Errors

0.60%

Error Rate

187 ms

Avg RT

612 ms

P95 RT

3984 ms

Max RT

13.82/s

Throughput

GO WITH CONDITIONS

SLA holds on 4/5 transactions; /payment p99 breach and 1.8% checkout errors require a canary rollout.

Production Confidence: 74.2/100 Risk Score: 25.8/100 Recommendation: Canary
10% for 60 min

SLA # Regression # Observability # Infrastructure # No critical findings

ROLLOUT PLAN

1. Canary 10% for 60 min with elevated dashboards
2. Ramp to 50% if p95 stays under 800 ms
3. Full rollout once error rate < 0.3%

ROLLBACK TRIGGERS

- checkout p95 > 1500 ms
- payment errors > 3%
- any CloudWatch alarm ALARM

HEALTH SCORES (0-100)

Performance

82

Infrastructure

88

Application

78

Database

84

Observability

92

Scalability

79

Reliability

86

Availability

95

Composite Perf

Release Readiness

Production Confidence

DETERMINISTIC FINDINGS

SEVERITY	RULE	TITLE	CONFIDENCE	EVIDENCE
HIGH	R-LATENCY	payment p99 breach	0.82	aggregate. per_transaction.POST /payment. rt_p99_ms=3984
MEDIUM	R-ERR-RATE	checkout error rate 3.1%	0.71	aggregate. per_transaction.POST /checkout.error_rate=0. 031
MEDIUM	R-TAIL	heavy tail on /payment (p99 = 4.5 x p95)	0.68	rt_p99_ms /rt_p95_ms=4.24

VISUAL ANALYTICS

```
.wf-label{font:12px -apple-system,Segoe UI,Roboto,sans-serif;fill:#334155}.wf-value{font:11px monospace;fill:#0f172a}.wf-title{font:13px -apple-system,sans-serif;fill:#0f172a;font-weight:600}Per-transaction latency (p50 / p95 / max, ms)POST /payment120 / 940 / 3984POST /checkout90 / 780 / 2140POST /cart55 / 380 / 950GET /product/{id}40 / 260 / 610GET /home15 / 110 / 240p50p95max

.sk-node{font:12px -apple-system,sans-serif;fill:#fff;font-weight:600}.sk-lbl{font:12px -apple-system,sans-serif;fill:#334155}.sk-title{font:13px -apple-system,sans-serif;fill:#0f172a;font-weight:600}Verdict # gate flowGO WITH CONDITIONS# No critical findings# SLA# Regression# Observability# Infrastructure
```

.dm-lbl{font:11px -apple-system,sans-serif;fill:#334155}.dm-title{font:13px -apple-system,sans-serif;fill:#0f172a;font-weight:600}Transaction dependency map (size = calls, colour = error rate)testGET /homeGET /product/{i...POST /cartPOST /checkoutPOST /payment

PER-TRANSACTION SUMMARY

TRANSACTION	COUNT	ERRORS	ERROR %	MIN	AVG	P95	MAX
GET /home	9824	0	0.00%	42 ms	55 ms	110 ms	240 ms
GET /product/{id}	6512	6	0.09%	88 ms	110 ms	260 ms	610 ms
POST /cart	3218	12	0.37%	120 ms	145 ms	380 ms	950 ms
POST /checkout	2641	82	3.10%	220 ms	280 ms	780 ms	2140 ms
POST /payment	2676	48	1.79%	260 ms	320 ms	940 ms	3984 ms

BASELINE DIFF

Previous run: 2026-02-04 14:22 UTC

- POST /payment rt_p95 up 18% vs previous baseline
- POST /checkout error_rate up +1.4pp

EXTERNAL OBSERVABILITY SOURCES

PROMETHEUS

QUERY	POINTS	MIN	AVG	MAX
cpu_saturation	4	0.620	0.668	0.710
request_error_rate	4	0.006	0.009	0.011

DATADOG

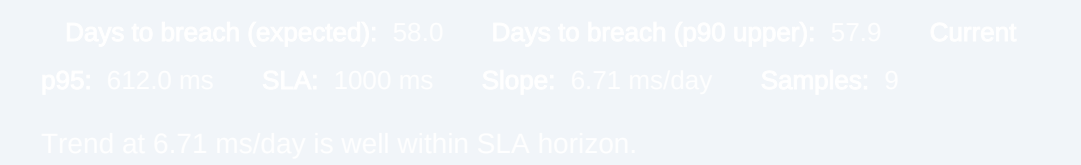
QUERY	POINTS	MIN	AVG	MAX
api_error_rate_pct	4	0.600	0.850	1.100
db_connection_saturation	3	0.520	0.567	0.630

DYNATRACE

QUERY	POINTS	MIN	AVG	MAX
payment_latency_ms	4	820.0	1038	1240

CAPACITY FORECAST

OK



BUSINESS IMPACT & CAPACITY FORECAST

Est. lost conversions: 84 Est. impact: \$3780 Peak supported TPS: 260 Cliff TPS: 320 Headroom (months): 4

Assuming \$45 AOV and current checkout drop-off.

AI ROOT CAUSE ANALYSIS

Provider: openai · Model: gpt-5.2

ROOT CAUSE HYPOTHESIS

Payment gateway upstream is the dominant contributor to p99 (4.5x p95 on POST /payment). Checkout errors correlate with the same latency spikes.

RECOMMENDED ACTIONS

1. Enable circuit breaker on payment client (2s open threshold).
2. Add idempotency retries on /checkout with jitter.
3. Scale API replicas to 6 during 15:00-19:00 UTC.

APPENDIX

Test: demo-checkout-load-2.0.3 **Env:** staging **App:** smart-shop **Started:** 2026-07-26 06:00:18 UTC **Stopped:** 2026-07-26 06:30:18 UTC

Raw evidence: local NDJSON, Splunk correlation JSON, O11y JSON, CloudWatch JSON, structured AI insights JSON, and CSV are all saved alongside this report inside the configured Output_Directory.

Generated by JMeter Smart Observability AI Plugin v2.0.3